

Md. Muhaiminul Islam (Jamee)

• Research Scholar • Embedded Systems Engineer • Deep Learning, IoT & Edge AI Specialist • Industrial Automation Expert • ROS Developer

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RESEARCH INTEREST

- Autonomous Vehicle System • Cyber Physical System • Edge AI & Embedded System • Computer Vision • Deep Learning
- Machine Learning • Robotics & Intelligent System • IoT • Control System

ACADEMIC QUALIFICATIONS

- **Rajshahi University of Engineering & Technology (RUET)** Rajshahi, Bangladesh
Jan 2020 – June 2025
 - B.Sc. (Engg.) in Electronics and Telecommunication Engineering
 - CGPA: 3.28/4.00 (1st Class 11th) (Last Semester GPA: 3.90; Ranked 3rd)** [View](#)
 - Thesis:** A Multi-Modal Robotic Framework for Real-Time Waste Detection and Categorization Using SLAM, IoT, and Convolutional Neural Networks.
- **Dhaka Residential Model College** Dhaka, Bangladesh
June 2017 – May 2019
 - Higher Secondary Certificate; **GPA: 5.00/5.00** [View](#)
- **Khulna Zilla School** Khulna, Bangladesh
Jan 2015 – Jan 2017
 - Secondary School Certificate; **GPA: 5.00/5.00** [View](#)

RESEARCH EXPERIENCE

- **Dept. of Electronics & Telecommunication Engineering, RUET** [View](#) Rajshahi, Bangladesh
July 2023 – Dec 2025
Undergraduate Research Assistant
- Designed an autonomous garbage detection and collection robot using Raspberry Pi, YOLOv5/Faster R-CNN-based object detection, and SLAM-based navigation with LiDAR-GPS fusion and Kalman filtering/A* path planning, achieving 97% detection accuracy for trash and 96% for non-trash items, with 97% precision and 98% recall.
- Engineered a GPS-GSM-based landmine and human-detection robot integrating Arduino, ESP32-CAM, and multi-sensor (PIR, gas, sonar, metal-detection) hardware, and an autonomous self-driving car using ROS and YOLO-based object recognition with real-time English-Bangla road-text translation, enabling remote hazard alerting and improved human-machine interaction across both platforms.
- Architected an AI-based smart notice board with a three-tier full-stack system (Python, Node.js, MySQL) for real-time facial identification and personalized content delivery, achieving 97% accuracy, precision, recall, and F1-score in facial recognition evaluation.
- Developed a real-time hand-gesture recognition system for human-machine interaction using deep learning-based classification models, achieving a 99.0% macro-F1 score.
- Investigated an advanced Fresnel algorithm for computer-generated holography to improve image quality in secure communication systems, optimizing Fresnel distance and operating wavelength (350–700 nm) to minimize ringing and blurring, and achieved a muxed-hologram entropy of 7.99 (near the ideal 8-bit maximum), confirming high-fidelity, low-correlation secure image reconstruction.

PUBLICATIONS (6 PUBLISHED, 2 IN PREPARATION)

All publications can be seen at [Google Scholar](#). The top 7 publications are provided as follows:

- **In Preparation:** A Systematic Benchmark of Edge AI Inference Frameworks for Resource-Constrained Autonomous Mobile Robot Platforms
- **In Preparation:** A Multi-Modal Robotic Framework for Real-Time Waste Detection and Categorization Using SLAM, IoT, and Convolutional Neural Networks.
- **M. M. Islam**, M. R. Hossain, F. Akter, M. F. Hossain and S. Datto, “Smart Waste Management Robot: Integrating IoT, Artificial Intelligence, SLAM, and CNN-Based Advanced Image Processing for Real-Time Trash Collection and Categorization,” in **2024 27th International Conference on Computer and Information Technology (ICCIT)**, 2024, pp. 3572-3577, doi: 10.1109/ICCIT64611.2024.11022551 [View](#)
- **M. M. Islam**, M. R. Hossain, F. Akter, B. K. Barman and S. Malakar, “The Design and Implementation of AI Based Smart Notice Board Using Advanced Machine Learning and Image Processing Techniques,” in **2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)**, 2024, pp. 1-6, doi: 10.1109/ICAEEE62219.2024.10561793 [View](#)
- **M. M. Islam**, F. Akter, M. R. Hossain, M. F. Hossain, S. M. Hasan and M. A. Y. Srizon, “An advanced iot and gps-gsm based real-time automated data monitoring robot with image processing and live streaming for human life and land mine detection systems,” in **2024 International Conference on Computer, Electrical & Communication Engineering (ICCECE)**, 2024, pp. 1-7, doi: 10.1109/ICCECE58645.2024.10497277 [View](#)
- M. A. I. Siddique, M. S. H. Shojol, **M. M. Islam**, S. Alam and O. Jyoti, “Smart border surveillance: Real-time cnn on drones with iot integration,” in **2023 26th International Conference on Computer and Information Technology (ICCIT)**, 2023, pp. 1-5, doi: 10.1109/ICCIT60459.2023.10441471 [View](#)
- **M. M. Islam**, M. R. Hossain and F. Akter, “Enhancing Regenerative Braking Efficiency in Electric Vehicles: An Advanced Fuzzy Logic-Based Control Strategy,” in **2023 IEEE 9th International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)**, 2023, pp. 263-267, doi: 10.1109/WIECON-ECE60392.2023.10456493 [View](#)

PROJECTS (25+ COMPLETED, INCLUDING 7 AWARD-WINNING IN COMPETITIONS)

The top 6 Projects are provided as follows. All Projects can be seen at my [Google Drive](#)

- **Four-Storey Elevator Control System** [View](#)
 - Designed a four-floor smart elevator using dual microcontrollers with an I²C-based master-slave architecture, separating decision-making, hardware control, and safety monitoring.
 - Implemented the SCAN elevator algorithm with DC/servo motors and synchronized door control to efficiently process floor requests while providing structured, direction-based operation.
 - Integrated ultrasonic and force sensors, dual LCDs, a 7-segment display, manual overrides, and external call inputs to enable real-time feedback, obstacle detection, overload protection, and enhanced operational safety.
- **Autonomous Self-Driving based Garbage Detection and Collection Robot using Deep Learning and Simultaneous Localization & Mapping** [View](#)
 - Developed an autonomous garbage detection and collection robot using Raspberry Pi, Pi Camera, and deep learning models including Fast R-CNN and YOLO for real-time waste-object detection.
 - Executed SLAM-based navigation by integrating LiDAR and sonar sensors with Kalman filtering and A* path planning to enable localization, obstacle avoidance, and efficient autonomous navigation.
 - Engineered the robotic motion and collection mechanisms using Arduino Mega, I²C communication, and servo motors, enabling autonomous detection, pickup, transport, and disposal of waste.
- **IoT and GPS-GSM Based Land Mine and Human Detection Robot with Automated Data Monitoring System and Image Processing** [View](#)
 - Created a multi-sensor robotic platform integrating Arduino, ESP32, ESP32-CAM, GPS-GSM, PIR, gas, sonar, and metal detection sensors for real-time landmine, human presence, and environmental hazard detection.
 - Executed GPS-GSM-based hazard alerting with live video streaming and image processing, enabling remote monitoring and automatic transmission of detected hazard locations to rescue teams via SMS.
 - Incorporated web and Android applications with Bluetooth-based control and motor drivers to support remote operation, real-time environmental monitoring, and safer deployment in hazardous areas.
- **Custom Real-Time Hand Gesture Recognition System for Edge AI Devices** [View](#)
 - Engineered an 8-command hand gesture recognition system using a 76-dimensional landmark feature extractor and 128-64 MLP architecture, training on 36,864 samples across 9 classes with augmentation and runtime rejection.
 - Validated the system on 3,631 public-test observations, achieving 98.8% known-command acceptance, 99.0% runtime macro-F1, and 0.4% unknown false-accept rate, substantially improving over the 92.9% baseline.
 - Optimized real-time deployment through ONNX inference, temporal gesture gating, and a geometry-based Dorsal fallback, correctly recognizing 331/331 Dorsal public-test samples while supporting live camera evaluation.
- **Autonomous Self-Driving Car with Text Translation** [View](#)
 - Engineered an autonomous self-driving car using ROS, Raspberry Pi, and Pi Camera, incorporating real-time lane detection and YOLO-based object recognition for intelligent navigation and obstacle awareness.
 - Applied Kalman filtering and Dijkstra's algorithm for vehicle localization and path planning, facilitating smooth, safe, and efficient navigation decisions.
 - Enabled real-time road-text detection and English-Bangla translation to interpret signs and boards, improving driver assistance and human-machine interaction.
- **AI Based Smart Notice Board** [View](#)
 - Architected an AI-powered smart notice board using machine learning, facial recognition, image processing, and socket communication to deliver personalized information in real time.
 - Engineered a three-tier full-stack architecture with HTML, CSS, JavaScript, Python, Node.js, and MySQL, enabling real-time facial identification, profile retrieval, and dynamic notice delivery through frontend-backend communication.
 - Achieved 97% accuracy, precision, recall, and F1-score in facial recognition evaluation, while automating personalized notice presentation based on recognized users and reducing manual information searching.

SKILLS SUMMARY

- **Programming:** C, C++, Embedded C, Python, SQL, MATLAB.
- **Libraries/Frameworks:** NumPy, Pandas, Matplotlib, OpenCV, Scikit-Learn, TensorFlow, PyTorch, Django, Flask, FastAPI, ONNX.
- **Software:** LaTeX, EasyEDA, KiCad, Altium Designer, Gazebo, CARLA, Huawei eNSP, CST Studio, Proteus, PSpice, EMU8086, COMSOL, Fusion 360, STM32 Cube (IDE, MX).
- **Hardware:** AVR MCU (ATmega328P, ATmega2560, ATmega32U4), ARM Cortex-M (STM32F103C8T6), NodeMCU (ESP8266), ESP32, JieLi (AC791, AC63), Rockchip RV1106, Raspberry Pi.
- **Communication Protocols:** TCP/IP, I2C, SPI, UART, RS232, CAN, MQTT, HTTP, LoRa.
- **Platforms & Tools:** Windows, Linux, ROS/ROS2, FreeRTOS, Zephyr RTOS, Git, GitHub, Docker, Google Cloud.

WORK EXPERIENCE

THT-Space Electrical Company LTD. [View](#)
Assistant Embedded System Engineer (R&D Team)

Nilphamari, Bangladesh
July 2026 – Present

- Engineered an 8-command hand gesture recognition system using a 76-dimensional landmark feature extractor and 128-64 MLP architecture, trained on 36,864 samples across 9 classes, and optimized with augmentation, runtime rejection, ONNX inference, and temporal gating to achieve 98.8% known-command acceptance, 99.0% macro-F1, and 0.4% unknown false-accept rate on 3,631 public-test observations.
- Designed and developed embedded firmware in C/C++ for microcontroller-based systems, including thermal and dot matrix printers, biometric IoT devices, and AI-powered embedded applications.
- Designed embedded hardware, mixed-signal circuits, schematics, and multilayer PCBs using Altium Designer and KiCad, including component selection, BOM optimization, prototyping, and hardware validation.
- Configured and optimized GPIO, Timers, PWM, ADC, UART, SPI, and I²C, while integrating sensors, actuators, wireless modules, and TinyML & Edge AI solutions for resource-constrained devices.

- Developed and deployed ML models and ROS using TensorFlow, Scikit-learn, PyTorch and Gazebo, integrating them into scalable embedded based applications with microPython; managed complete TinyML workflows including model training, data preprocessing, API deployment, and edge integration.
- Built different types of robotics structure using Node MCU, ESP32 Cam, Arduino & Raspberry Pi.
- Led and mentored a cross-functional development team; managed full project lifecycles and aligned technical solutions with business objectives through collaboration with executives and stakeholders.

HONORS AND AWARDS (30+ NATIONAL & INTERNATIONAL AWARDS)

All achievements can be seen at [Google Drive](#)

- **Technical Scholarship by University Grants Commission, Bangladesh** [View](#) 2021 | 2022 |
Organized by: Dept. of ETE, RUET 2023 | 2024
- **Top 5 Best Project Award on course ETE 2200: Electronic Project Design & Development** [View](#) 2022
Organized by: Dept. of ETE, RUET
- **Technical Award Winner in Creative Category at International Robot Olympiad (Among 30+ Countries)** [View](#) 2020
Organized by: International Robot Olympiad Committee (IROC)
- **Bronze Medalist in Creative Category at Bangladesh Robot Olympiad (Among 1200+ Students)** [View](#) 2020
Organized By: Bangladesh Robot Olympiad Committee (BdROC)
- **National Champion at CISCO IoT Hackathon (Among 30+ Colleges & Universities)** [View](#) 2018
Organized By: CISCO
- **Divisional Champion & Best Meritorious Student Award in Math (Among 1500+ Students)** [View](#) 2015 | 2016 | 2018
Organized By: Creative Talent Hunt Competition by People's Republic of Bangladesh
- **Second Runner-up in Bangladesh Mathematical Olympiad (Among 1000+ Students)** [View](#) 2014 | 2016 |
Organized by: Bangladesh Mathematical Olympiad Committee (BdMOC) 2017 | 2018
- **First Runner-up in UIU-Venturas First Robotics Competition for Colleges (Among 20+ Colleges)** [View](#) 2018
Organized by: United International University & Venturas
- **National Champion in National High School Programming Contest (Among 2000+ Students)** [View](#) 2015 | 2017
Organized by: ICT Division & National High School Programming Contest Committee, Bangladesh (NHSPC)
- **7th at Bangladesh Physics Olympiad (Among 600+ Students)** [View](#) 2014 | 2017
Organized by: Bangladesh Physics Olympiad Committee (BdPhOC)
- **Champion at Bangladesh Junior Science Olympiad (Among 1000+ Students)** [View](#) 2017
Organized by: Bangladesh Junior Science Olympiad Committee (BdJSOC)
- **6th at Bangladesh Science Olympiad (Among 1500+ Students)** [View](#) 2016
Organized by: Bangladesh Science Olympiad Committee (BdSOC)
- **Divisional Champion at 36th & 37th Bangladesh National Science Fair (Among 40+ Schools)** [View](#) 2015 | 2016
Organized by: Ministry of Science and Technology, Bangladesh
- **Divisional Champion at National Science Carnival, Bangladesh (Among 1000+ Students)** [View](#) 2015 | 2016
Organized by: Ministry of Science and Technology, Bangladesh

CO-CURRICULAR EXPERIENCE

- *Research & Planning Secretary* at **Astronomy & Science Society of RUET**, Rajshahi, Bangladesh Jun 2023 – Jun 2024
- *Academic Team Member* at **Bangladesh Robot Olympiad (R&D Team)**, Dhaka, Bangladesh May 2021– May 2024
- *Academic Team Member* at **Bangladesh Mathematical Olympiad (R&D Team)**, Dhaka, Bangladesh Feb 2021– Dec 2024

CERTIFICATIONS

- Machine Learning & Data Science Bootcamp (Starter Pack) by Code Studio Academy [View](#) Feb 2023 – July 2023
- PCB Design & Fabrication Workshop by Dept. of ETE, RUET [View](#) Aug 2022 – Aug 2022
- Fundamentals of Internet of Things Workshop by RUET IoT Club [View](#) July 2022 – July 2022
- Workshop on Advanced Softwares for ICT by IICT, RUET [View](#) Feb 2021 – Feb 2021
- Basic Arduino Programming Course by Think-Robo [View](#) July 2020 – Aug 2020

REFERENCES

- 1. Md. Rakib Hossain**
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